STAFF REVIEW · H446 240 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Requested result is not already stored.

2. Only when both A and B are true.

3. 0.

4. Yes, because seconds>=1 is true.

I Do — source worked steps

1. Predict first — The valid endpoints were incorrectly rejected.

2. Model and check — Change the operators to >= and <=, and test −1 and 11 as invalid values.

We Do — guided CHECK

1. Predict first
OR accepts 61 because only one side needs to be true.
Model and check
AND requires both limits to hold and correctly rejects 61.

You Do — source mark schemes

1. [4] Use the lower inclusive boundary x >= 5 [1] and upper inclusive boundary x <= 15 [1] together with AND: x >= 5 && x <= 15 ; x=5 is accepted [1] and x=15 is accepted [1]. Both clauses must be combined in the final condition.

2. [6] 100>=12 is true [1]; use && [1]; 11 false [1], 12 true [1], 18 true [1], 19 false [1].

3. [6] Reproduce failing case [1]; identify faulty condition [1]; correct code [1]; retest valid boundaries [1] and invalid neighbours [1]; retain automated regression tests [1].

Exit Ticket — source answer

1. False. OR often wrongly accepts values outside the range.